Project Deliverable F: **Comprehensive Prototyping and Testing**

GNG 1103 – Engineering Design

Faculty of Engineering – University of Ottawa

Objectives:

- Reflect on feedback so far.
- Develop a comprehensive prototype according to the test plan.
- Conduct testing and analysis against target specifications.
- Devise a test plan for an improved comprehensive prototype.
- Identify comprehensive prototyping and testing tasks, estimate their duration and assign owners.

Instructions:

Teams will complete the appropriate sections of the deliverable template (GNG1103_Deliverable-F_Template.docx), the BOM (GNG1103_BOM_Template.xlsx), and the test plan (GNG1103_TestPlan_Template.xlsx) available on Brightspace, as described in the deliverable requirements below:

	Required Elements	**Evaluation Rubric**	✓
1.	Title page, table of contents, bibliography	• The title page has been filled in with all appropriate info. • Automatically generated table of contents has been updated to reflect document content using built in headings. • Real citations have been added where expected when referencing external information using the IEEE format. • All citations referenced in text are included in the bibliography using the IEEE format.	
2.	Introduction	• Introduction section completed, providing the problem statement and enough context for a non-technical reader to understand your project. • Updated from the previous deliverable, as required.	
3.	Feedback	• Feedback received from users you have interviewed and the client on the focused prototypes and testing results is described. • How this feedback will be used to inform and improve your future design work is specified.	

4.	Comprehensive prototype	• Detailed visual evidence of the comprehensive prototype your team developed, as you described in the previous deliverable, is provided.	
5.	Comprehensive test plan	• The test plan developed in the previous deliverable is included, updated as necessary. • Results from the tests conducted (one per team member) with the prototype are captured and interpreted. • The test plan is submitted as an Excel document.	
6.	Target specifications	• Target specifications from the previous deliverables are included. • Target specifications have been updated to account for any new information or insight (if necessary).	
7.	Analysis	• A simple analysis of critical prototype components or systems is performed, based on your current engineering science or other knowledge. • Results of the analysis are compared to the target specifications and a discussion is provided on prototype performance.	
8.	Comprehensive test plan	• A description of how the comprehensive prototype will be improved (or to make another) before Design Day is provided. • A test plan is developed, using the template, including the following elements: • A number of tests are identified, equal or greater to the number of members in your team, to verify assumptions. • An objective (why) is identified for each test. • A test description (what) is provided for each test. • An analysis method (how and when) is described for each test. • How the improved comprehensive prototype will be used for each test is described. • The test plan is submitted as an Excel document.	
9.	Bill of materials (BOM): comprehensive prototype	• Using the template, an updated BOM is provided, including all additional parts, components, software, and materials (as required) used to improve your comprehensive prototype. • The BOM is submitted as an Excel document.	

10.	Task and time estimates	• Using Trello, or a TA approved alternative, a list of all tasks that must be completed before design day is provided. • A time estimate for the duration of each task is provided. • A team member is identified as responsible for each task.	
11.	Conclusion	• Conclusion section completed, providing reasonable and appropriate concluding remarks about important insights, current status of the project, and outstanding work remaining.	
12.	Text quality	• All text is well written, clear, concise, and has been checked for grammar and syntax at a level that can be reasonably expected when using writing aid tools.	

Design Day:

You will demonstrate the final version of your comprehensive prototype on Design Day. See the Design Day deliverable for more information.

Prototyping and Budget Information:

Two prototyping deliverables will be due from now until design day (see due dates in Brightspace). The first deliverable will focus on developing low fidelity focused prototypes (one per team member) to test assumptions. Typical objectives include communicating and getting feedback for ideas, verifying feasibility, analysing critical mechanisms, circuitry or code blocks, and reducing risk and uncertainty. These prototypes should be made using materials and components that cost very little (e.g. things found around the house, scraps, etc.) or using modelling and simulation software. A simple analysis of critical components or systems based on your current knowledge of engineering science or other material should also be included. The second prototyping deliverable will focus on developing a fully functional comprehensive prototype. You will receive a budget for this (see below). Many successful groups do *more* prototyping and testing than is asked for in the deliverables, based on their specific project risks. Prototyping tips are provided here:
https://wiki.makerepo.com/wiki/Design_for_manufacturing

For prototyping and testing, each team will be allocated a total of $25-$100 (see your project description on makerepo for the correct amount). You can take the opportunity to concentrate your purchases on one subsystem and collaborate with other teams to share resources. For software-based projects, teams are encouraged to use free tools when possible. To spend your budget, you must have your BOM approved by your TA ahead of time. You will purchase materials and components yourselves and you will bring your TA the original receipts for a reimbursement. It is easier if only one person on your team makes all the purchases, but you

must provide receipts for all purchases. See the purchasing and reimbursement guide: https://wiki.makerepo.com/wiki/Purchasing_Guide.

Submission:

Each team (***only one person from each team***) must submit PDF copies of the deliverable template, BOM template, and test plan template, ***containing all required elements*** of this deliverable by uploading the files as a single submission into Brightspace.

Grading:

Each required element of this submission will be graded pass/fail based on whether you have reasonably included all aspects, as stated above.

Due Date: See Brightspace.